

Designing an Eight-Bit Arithmetic Logic Unit Using ModelSim



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This project describes the designing of an 8-bit arithmetic logic unit (ALU) using Verilog programming language. It includes writing, compiling and simulating Verilog code in ModelSim on a Windows platform.

TABLE I
Pin Description of the ALU

Pin	Pin description
<A>	8-bit input signal A
	8-bit input signal B
<SEL>	6-bit selection signal
<S>	8-bit output signal
<O>	Output carry signal

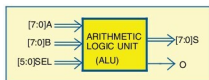


Fig. 1: Pin diagram of the ALU

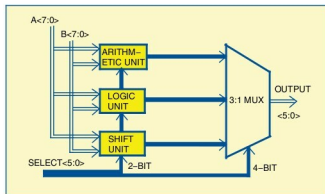


Fig. 2: ALU architecture

In digital electronics, an ALU is a digital circuit that performs arithmetic and bit-wise logical operations on integer binary numbers. It is a fundamental building block of the

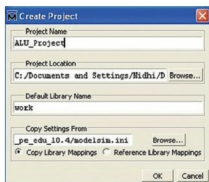


Fig. 3: Create Project window



Fig. 4: Add items to the Project window

central processing unit (CPU) found in many computers and microcontrollers.

Inputs to an ALU are the data to be operated on (called operands) and a code indicating the operation to be performed, while the

TABLE II
ALU Functions Table

SEL	Function
<i>Arithmetic unit</i>	
0000	a+b
0001	a+b+1
0010	a+b-1
0011	a-b
0100	a-b-1
0101	a-b+1
0110	a==b
0111	a<=b
1000	a!=b
1001	a-1
1010	b-1
<i>Logical unit</i>	
0000	~a
0001	~b
0010	a & b
0011	~a & b
0100	a & (~b)
0101	~a & (~b)
0110	a b
0111	~a b
1000	a (~b)
1001	~a (~b)
1010	a ^ b
1011	~a ^ (~b)
1100	~(a & b)
1101	~(a b)
<i>Shift unit</i>	
0000	a<<1
0001	a>>1
0010	a>>1, out[0]=a[0]
0011	a<<1, out[7]=a[7]
0100	a>>1, out[0]=a[7]
0101	a<<1, out[7]=a[0]
0110	b<<1
0111	b>>1
1000	b>>1, out[0]=b[0]
1001	b<<1, out[7]=b[7]
1010	b>>1, out[0]=b[7]
1011	b<<1, out[7]=b[0]